

Lorenzo Maggi

Biomedical Engineer

Hands-on biomedical engineer (PhD) with experience in **surgical robotics, knee biomechanics, and ex-vivo experimentation.** During my PhD I developed a knee kinematic simulator and contributed to the design of **Robin®**, a robotic system for total knee arthroplasty.
Strong **maker** mindset **(mechanics, electronics, 3D printing)** and a focus on **translational** research.

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https://lorenzo-maggi.github.io/

Languages

Italian (mother tongue)
English (fluent)
Spanish (beginner)

Core skills

Robotics & control: **KUKA LBR Med**;
Software/Coding: **SolidWorks, Office Suite, Python, C/C++, Java, MATLAB, Arduino**.

Lab skills: **sensor** calibration **soldering** (incl. SMD), **3D printing** (FDM), general **mechanical skills**.

Licence of professional engineer.
Scientific writing.
Knee biomechanics.
Driving licence: A, B.

Hobbies & Other activities

Volunteering**:**

- **Ambulance driver** and first **responder** (2014 - present);
- **BLSD and safety trainer** for internal and external staff;
- Event **organizer** and **cook** (with HACCP) for **country fair** (**Fa'Festa** 2011 - present).

Personal projects:

- **DIY sous-vide cooker**;
- **Programmable PSU**
- **Full restoration** of a **Piaggio Si**.

Interests:

- **Trekking and mountain biking;**
- **Cooking**.

Experience & Education

R&D Spine Implants Engineer
September 2026 - ongoing
Clover Orthopedics Srl

- **Mechanical design** and development of **spinal implants**;
- **MDR oriented** design and documentation review**.**

R&D Robotics and Biomedical engineer
November 2025 - August 2026
Orthokey Italia Srl

- **Co-development** of **Robin®** robotic system for **TKA**;
- **ISO 13485, IEC 60601, IEC 80601** oriented design;
- **KUKA LBR Med** software development;
- Collaboration with **National Research Council** for Murine emotional analysis via **neural networks**.

PhD in Industrial Engineering (cum Laude)
November 2022 - October 2025
Orthokey Italia Srl - University of Florence

- **Development** of an innovative **knee laxity** assessment robotic tool based on **KUKA LBR Med**;
- **Development** of a **force-feedback control** for rehabilitation **hand exoskeleton**;
- **2 years** of Industrial experience at **Orthokey Italia Srl**;
- A brief **summary** of my **PhD**: https://youtu.be/5DqW-fMwWBU

Visiting PhD researcher
September 2024 - March 2025
Université Libre de Bruxelles (Prof. B. Innocenti)

- **6-month abroad experience**;
- **Development** of a portable **knee kinematic** simulator for kinematic analysis of different **TKA implants**;
- Implementation of a **PID controller** to simulate **quadriceps tendon loading**;
- Participation in several **cadaveric knee tests** .

MSc in Biomedical engineering (110/110 cum Laude)
October 2022

BSc in Mechanical engineering
February 2021
University of Florence

I hereby authorize the processing of my personal data in accordance with Legislative Decree No. 196 of June 30, 2003 and Article 13 of the EU General Data Protection Regulation (GDPR).